

A Study of Gender Recognition From Iris: A Literature Survey

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Abstract—Recent biometric research has examined the possibility of obtaining attributes such as hair color, age, gender, weight, ethnicity, height, etc. from biometric traits, face, hand geometry, fingerprints and iris. This paper examines detailed study about different process, for each step of predicting gender from iris for achieving better accuracy and authentication. Capturing the iris image in high quality specification camera is used to achieving specific features of iris. Different algorithms and software systems are used to locate the boundaries of an iris image. Mutual Information(MI) is better used to compare other feature in geometry, texture etc. UND_V and GFI datasets are used to get more accuracy in SVM classifier for better gender prediction.

Keyword—Image processing, Feature Selection, Mutual Information, Gender Classification.

I. INTRODUCTION

In biometric there are different recognition methods are occurring such as finger print, face, DNA, ears, iris, etc. Each of these biometric has them on characteristics. For iris recognition, that have a unique characters compared to other biometric. The characters are,

- Random pattern of great complexity and uniqueness.
- Stable features from age three to whole life.
- Highly protected, internal organ of the eye.
- Distance up to some meters the eyes are externally visible.
- The iris of two eye in an individual contain completely independent pattern.
- Even though the identical twins possess uncorrelated iris pattern.

High-resolution and distortion free images are taken for iris pattern recognition of human eye. The pattern recognition takes different algorithms that's are Classification algorithms (It is supervised algorithms to predicting categorical labels

based on parametric or non-parametric.), Clustering algorithms (Is unsupervised algorithms to predict categorical labels.), Regression algorithms (To predict real value labels and These are supervised and unsupervised), Multilinear subspace learning algorithms (To predicting labels on real value), Sequence labelling algorithms (Predicting sequence of clear-cut labels).

Iris recognition is used in different application areas like Immigration, border control, healthcare, welfare, finance, banking, hospitality, tourism, public safety, etc.

In future, gender recognition is becoming a fundamental step in many human interaction applications. Different approaches are used for gender recognition. Gender recognition by face in different characteristic methods such as Gabor filters, PCA and LBP, COSFIRE filters, deep convolution network and Eigen features. The other gender recognition by using mobile biometric data, skin texture feature, whole body image, speaking nature. The distinctive features of male and female subjects are the hairline of males are higher and that of female lower. In male they have usually thick and under the orbital rim as in case of female sits higher and more arched. Generally female eyes are appearing in large and male are small and distance between eye and eyebrows are small for male and large in female.

The following are basic processing stages of iris based gender prediction. First an acquisition step. It involves how eye image is captured. The next step is segmentation. It localizes the iris region from the acquired image. This step involves finding the boundary of pupil. The feature extraction step extracts the geometric features or texture features or any other features. The next stage is training, here these feature parameters are trained according to the configuration required. In prediction step the data considered at the output of the previous step and performs the gender classification.

II. RELATED WORKS

Iris recognition algorithm is first created by John G. Daugman. Iris recognition system take high resolution images of the iris of the person's eye.

A number of studies are occurring in gender recognition from face images [1][2], however only limited studies in gender prediction from the fingerprint [3][4] and eye images [5][6]. Also, there have a limited number of studies have been reported for gender prediction from ocular biometric in visible spectrum [6][7].

III. CAPTURING IRIS IMAGE

Firstly, talking about the iris images for recognition purpose, there are two types: Visible wavelength iris images and Near infrared images. Figure 1 shows the VW and NIR images. The human eye appearing richly structured and less visible wavelength band. Near infrared spectrum enables the blocking of corneal mirror like reflections from the bright encircling environment. This mechanism allowing wavelength from narrow band illuminator back into the iris camera.

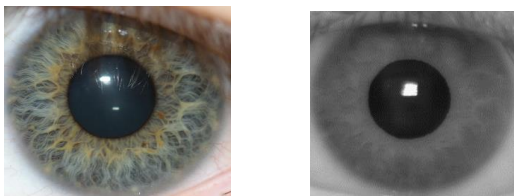


Figure 1: Visible Wavelength iris image & NIR iris image.

Coloring component of the iris is pigment (known as a chromosphere). It comprises two clear heterogeneous macromolecules. One is eumelanin (brown – black) and other are phenomelanin (yellow – reddish) [8]. Transmission density at longer wavelength in Near infrared spectrum is negligible and shorter wavelength within visible wavelength can return rich pattern.

A. Iridian LG EOU 200 System

The system has three infrared light emitting diodes. One at top, one at the bottom left and one at the bottom right. The distance between eye and the system in three stages with each of the LED lights getting activated one after another. This iris image has a resolution of 640x640 and intensity value range is an 8bit range. Still iris images and video iris images are available in LG2200 sensor.

B. I-Scan 2

The device is a dual iris capture scanner, built in U.S. It works in total darkness and in bright sunlight. This device is a NIR illumination in 700 – 900nm. Gives the bitmap images at maximum rate of 7.5fps. Capture rate is 15,30fps and operating system is windows, hardware is 1.3Ghz or higher Pentium 4.

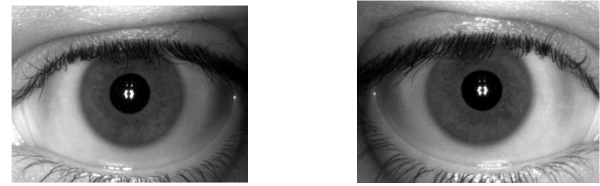


Figure 2: Left and Right image of I-Scan2 device

C. CCD Camera

CCD is developed up into a large number of small light sensitive areas. These sensitive areas can be used to build up an image of the scene of interest. CCD camera in iris sensor for capturing NIR images. Wavelength ranging about 1050nm with a peak sensitivity at round 700nm. Distance between eye to camera is 3 to 10 inches. Computer locate centre of a pupil and iris, edge of the pupil and iris, then eyelids and eyelashes, When the camera takes a picture.

The pupil trans out black allowing the software to locate the iris. Irises are unchanging and is a protected structure so scanning is very accurate.

IV. SEGMENTATION

A. ND Iris Software

The ND-IRIS software [8] deploys the segmentation by canny edge detector of Circular Hough Transformation. This software, firstly detect the limbic boundary then the pupillary boundary.

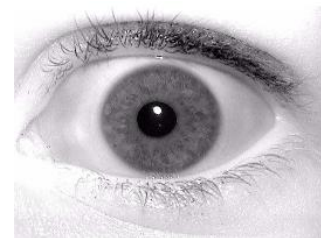


Figure 3: Image from ND-iris system

The limbic boundary test is to check how much the iris is darker compared to sclera. This check through with small portion on the left and right sides of the sclera iris boundary.



Figure 4: Detecting Pupil and Limbic boundary

The pupil boundary test explain how much pupil is darker.

This check is done with a small region on the left and right sides of the candidate boundary with the assumption is the iris image is oriented right side up. After the boundaries detection the software, again split the iris area into four equally sized parts. Left top, Left bottom, Right top and Right bottom. In these separation there have an overlap of half of the pupil radius between these four window. Connect each of these four windows and get a result of togetherness.

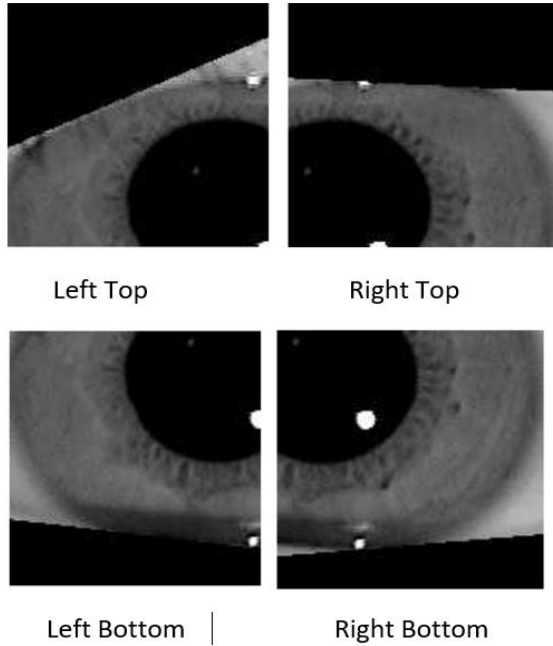


Figure 5: ND-iris 4 window

B. Wildes' System

In Wildes' System [8], the gradient based Hough Transform is used to determine the circular boundaries of an iris. This software firstly image intensity information is converted into a binary edge map. The binary edge map is generated by using Gaussian filter or gradient based edge detection. Then circular Hough space vote is analysed to estimate the three parameters of one circle (X_c, Y_c, r_c).

C. Daugman's Algorithm

Daugmans's algorithm base is Integro Differential Operator.

$$\max(r, x_0, y_0) \left| G_{\sigma}(r) * \frac{\partial}{\partial r_{r, x_0, y_0}} \oint \frac{-1(x, y)}{2\pi r} ds \right| \quad (1)$$

This equation is done twice, one for finding the contour of limbus contour and other for finding pupil contour. The Daugman uses a threshold value, if the finding value is greater than the threshold then ignores it. This threshold method is used for neglecting the difference occur in dark and bright spot

illumination.

V. FUTURE EXTRACTION

When the iris region is segmented from the eye image, then it is transform into fixed dimensions for allowing the comparisons. They have large number of methods like, Gabor filter, Wavelet transform, Hilbert transform etc. are used to extract features from normalized iris image.

A. Geometric Features

The geometric feature means that, the pupil and limbus segmentation stage variables are extracted. The variable and description are explained in following table.

Variable	Description
i_x	The centre of iris in x coordinates.
i_y	The centre of iris in y coordinates.
P_x	The centre of pupil in x coordinates.
P_y	The centre of pupil y coordinates.
i_r	Radius of the iris.
P_r	Radius of the pupil.
distCenter	Distance between centre of pupil and centre of iris.
irisArea	Total area of the iris () and pupil ().
pupilArea	The pupil area calculated using pupil radius.
diffArea	Difference between iris area and pupil area this gives the true iris area.
areaRatio	Ratio of total area of iris to total area of pupil.

Table 1: Geographical feature variable

B. Texture Features

As like in geographical feature extraction, texture features are extracted from the segmentation and encoding process.

Variable	Description
XRealMean	The mean value of the real component of the complex number in row x.
XRealStd	The standard deviation of the real component of the complex numbers in row x.
XRealVar	The variance of the real

	component of the complex numbers in row x.
YRealMean	The mean value of the real component of the complex number in row y.
YRealStd	The standard deviation of the real component of the complex numbers in row y.
YRealVar	The variance of the real component of the complex numbers in row y.
colVar	Overall measure of variation of the real components of the complex numbers in the entire array.

Table 2: Texture Feature variable

C. Statistical Features

This has more features from the iris image. The statistical features are computed along angular direction and radial direction. The radial direction movement is that the movement in iris pupil boundary to iris-sclera boundary. The statistical features calculate the mean, median and standard deviation.

Variable	Description
$\bar{X}_r$	Mean computed along each row.
M_{dr}	Median computed along each row.
S_r	Standard deviation computed along each row.
$\bar{X}_c$	Mean computed along each column.
M_{dc}	Median computed along each column.
S_c	Standard deviation computed along each column.

Table 3: Statistical Feature Variable

VI. CLASSIFICATION

For the classification process they have many classifier algorithms are used. Each paper of gender recognition explains one of these classification algorithms. The classification algorithms are C4.5 Decision Tree, SVM, Multi-Layer Perception, Optimized Incremental Reduced Error Pursing etc.

The C4.5 decision tree algorithm is the core learning technique. It can reach accuracies close to 80%. In the case of SVM and MLP, the accuracies are close to 80% in SVM and generate greater accuracy in MLP. The accuracies are depending on which feature extraction method are choosing. The optimized incremental reduced error pursing method get the accuracy less than in comparing MLP.

VII. CONCLUSION

This paper, investigate different steps for gender classification from iris. Each step has detailed study of the different approaches.

Number of features are explained in this paper for predicting gender accurately. Here explained four feature selection method, the statistical and 2D- DWT are providing much more feature.

The feature selection method MI are the newest approach. mRMR and CMIM [9] are using pairs of feature and improved methods W-mRMR and W-CMIM [10]. By using this four measures got the best accuracy 85.33% and 84.33% on the left iris and right iris code experiment.

REFERENCES

- [1] T. Sarna, P. Meredith. "The chemical and physical properties of eumelanin". *Pigment Cell Res.* 19 (6): 572–94. PMID 17083485. doi:10.1111/j.1600-0749.2006.00345. x.
- [2] A. M. Burton, V. Bruce, and N. Dench. What's the difference between men and women? evidence from facial measurement. *Perception*, 22:153–176, 1993.
- [3] D. Cao, C. Chen, M. Piccirilli, D. Adjero, Bourlai. T, and Ross. A. Can facial metrology predict gender? In *IJCB*, pages 1–8, Oct 2011.
- [4] Mark A Acree. Is there a gender difference in fingerprint ridge density? *Forensic Science International*, 102(1):35 – 44, 1999.
- [5] S. Gungadin. Sex determination from fingerprint ridge density. *Internet Journal of Medical Update*, 2:4–7, 2007.
- [6] V. Thomas, N. Chawla, K.W. Bowyer, and P.J. Flynn. Learning to predict gender from iris images. In *2015 International Conference of the Biometrics Special Interest Group (BIOSIG)*, Sept 2007.
- [7] Y. Dong and D. L. Woodard. A feasibility study: Eyebrow shape-based features for gender classification. In *2011 IJCB*, pages 1 – 8, 2011.
- [8] X. Liu, Optimizations in Iris Recognition. PhD thesis, Notre Dame university, 2006.
- [9] J. Tapia and C. Perez, "Gender classification from the same iris code used for recognition," *IEEE Transactions on Information Forensics and Security*, vol. 8, 2015.
- [10] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information criteria of max-dependency, max-relevance, and minredundancy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 27, no. 8, pp. 1226–1238, 2005.